T1 According to the title, we can set up constraints based on the problems conditions:

(a) $x_j \le M y_j$

(b) $y_1 \le y_2 + y_3$.

(c) $x_1 \ge 100 y_1$

(d) $x_1 \ge 5(y_2 + y_3)$.

(e) $y_2 \ge y_1$

$x_j \ge 0$, $y_j \in \{0, 1\}$, $j = 1, 2, 3$

T2 According to the title, we can transform this problem into a LP problem:

$$\text{maximize } Z = 5x_1 + 2x_2.$$

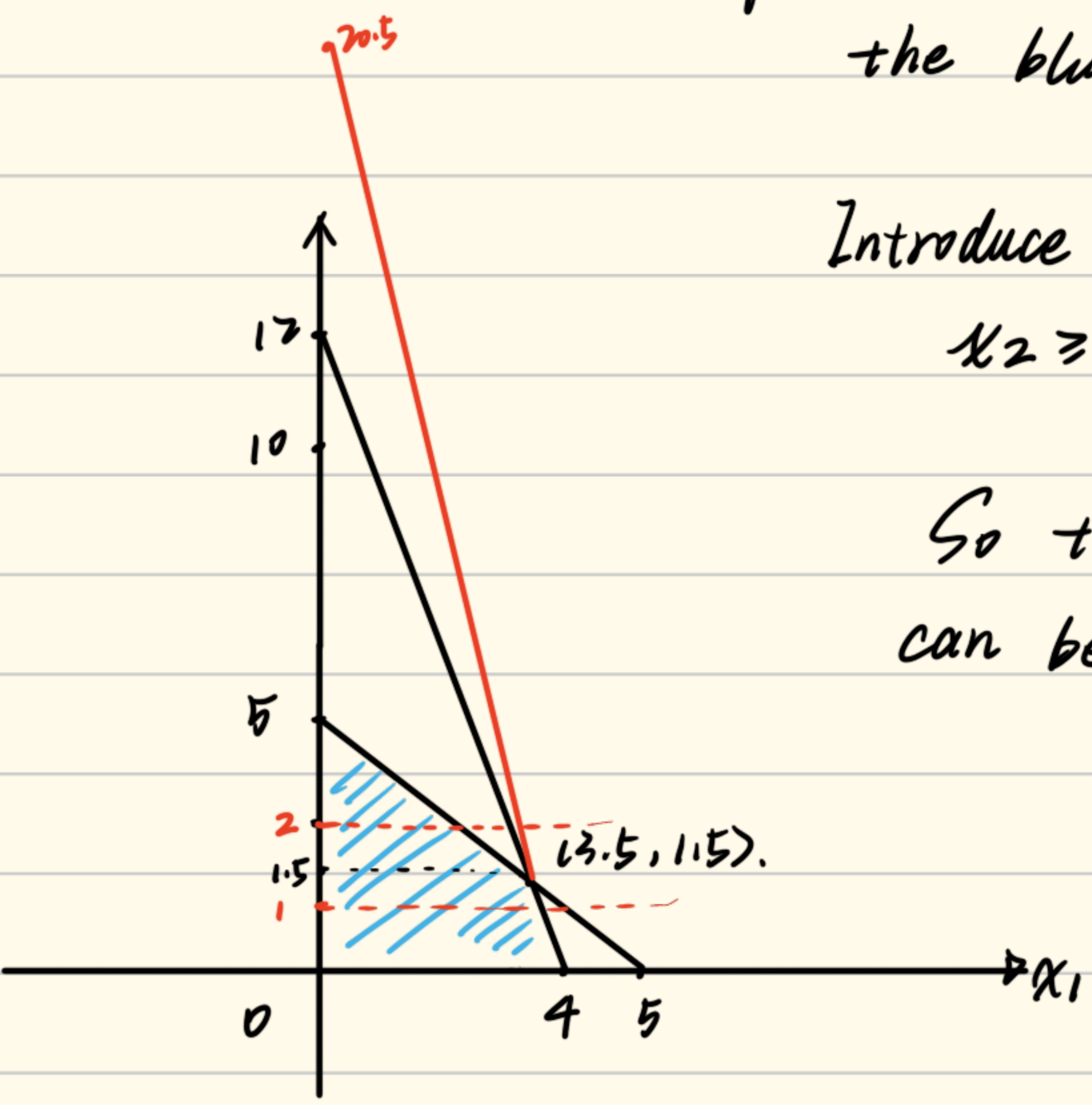
$$\text{s.t. } 3x_1 + x_2 \le 12$$

$$x_1 + x_2 \le 5$$

$$x_1, x_2 \ge 0.$$

So we can express this problem in diagram:

the blue region is the feasible region.



Introduce two constraints:

$x_2 \ge 2$; $x_2 \le 1$.

So the original problem can be divided into 2 parts:

Part 1:

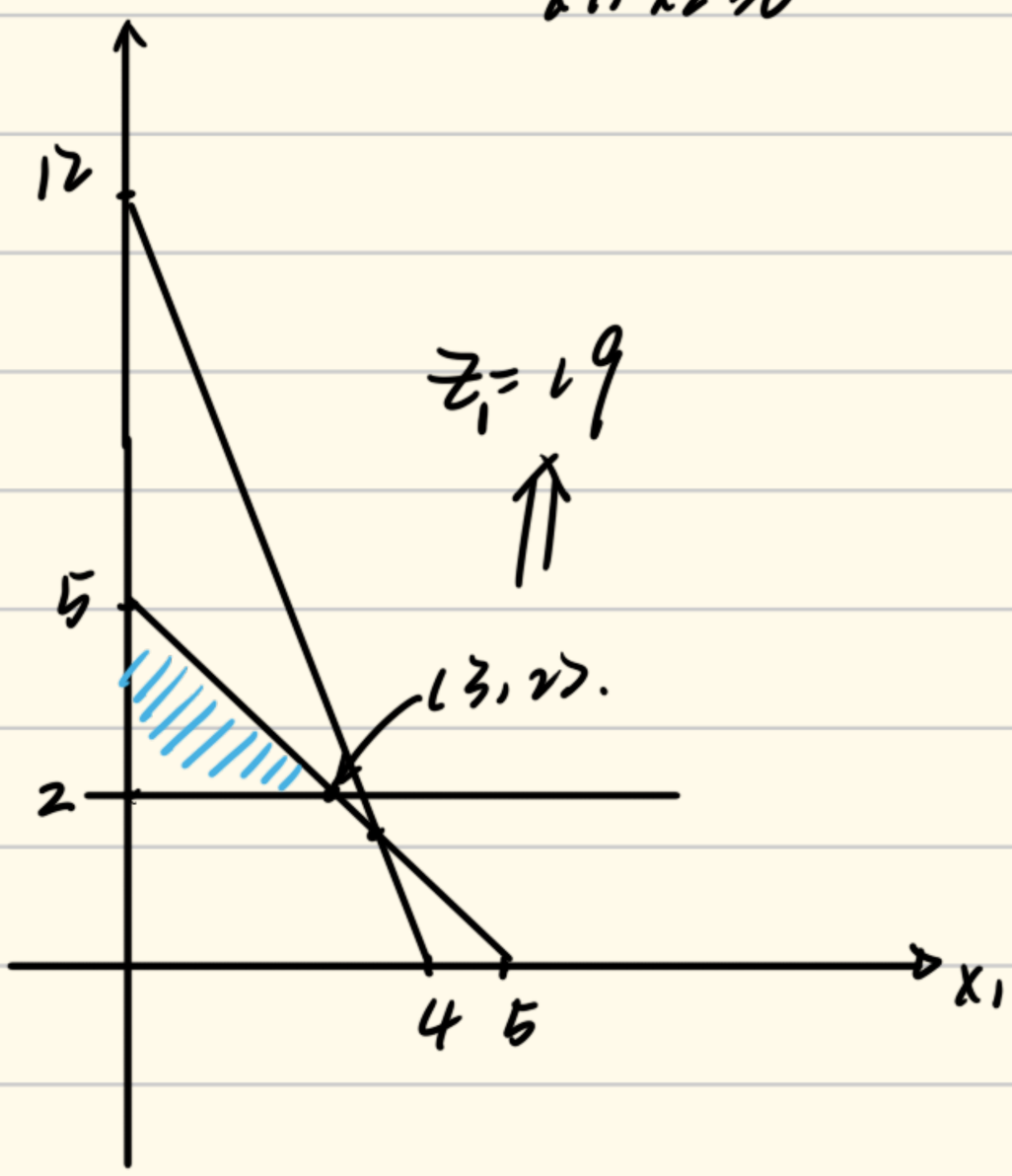
maximize $Z = 5x_1 + 2x_2$.

s.t. $3x_1 + x_2 \le 12$

$x_1 + x_2 \le 5$

$x_2 \ge 2$

$x_1, x_2 \ge 0$



Part 2:

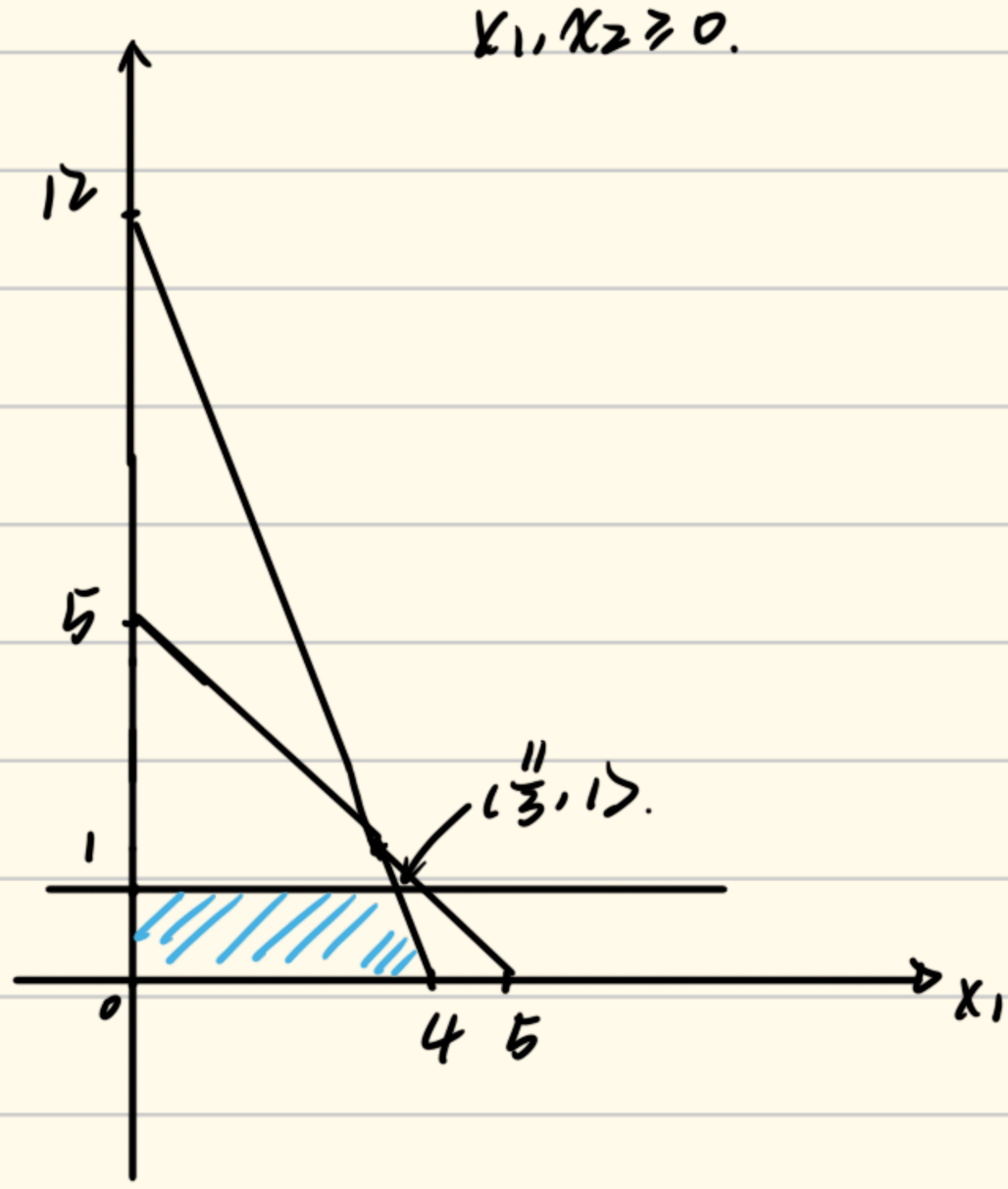
maximize $Z = 5x_1 + 2x_2$

s.t. $3x_1 + x_2 \le 12$

$x_1 + x_2 \le 5$

$x_2 \le 1$

$x_1, x_2 \ge 0$.



Because of Part 2's optimal solution is NOT integer, we need keep branching:

Part $2^{(1)}$:

maximize $Z = 5x_1 + 2x_2$

s.t. $3x_1 + x_2 \le 12$

$x_1 + x_2 \le 5$

$x_2 \le 1$

$x_1 \le 3$

$x_1, x_2 \ge 0$.

Part $2^{(2)}$:

maximize $Z = 5x_1 + 2x_2$

s.t. $3x_1 + x_2 \le 12$

$x_1 + x_2 \le 5$

$x_2 \le 1$

$x_1 \ge 4$

$x_1, x_2 \ge 0$.

We can solve that the optimal solution of Part $2^{(1)}$、Part $2^{(2)}$ is: (3, 1)、(4, 0)

$\Rightarrow Z_2^{(1)} = 17$, $Z_2^{(2)} = 20$

$\because Z_2^{(2)} > Z_1 > Z_2^{(1)}$

$\therefore$ (4, 0) is the optimal integer solution, $Z = 20$.